

Exercise No. 3

STATISTICAL TESTS AND PSEUDORANDOM NUMBERS GENERATORS

Adam Cherek, s207250

3.1

Plain text:

Perfect pancakes are easier to make than you think. This pancake recipe produces thick, fluffy, and all-around delicious pancakes with just a few ingredients that are probably already in your kitchen (and it's so much better than the boxed stuff). You likely already have everything you need to make this pancake recipe.

$\alpha = 0,05$	Frequency test	Serial test	Poker test	Runs test
Plain text	Fail	Fail	Fail	Fail
Caesar cipher	Fail	Fail	Fail	Fail
Hill cipher	Fail	Pass	Fail	Fail
DES (CBC)	Pass	Pass	Pass	Fail
AES (CBC)	Pass	Pass	Pass	Pass

Conclusions:

- Plaintext & Caesar Cipher: Standard English text is highly non-random (ASCII bit representations contain heavy biases, e.g., leading zeros). The Caesar cipher is a monoalphabetic substitution cipher; it merely shifts characters but preserves the exact statistical frequency of the plaintext bitstream. Therefore, both fail all statistical tests completely.
- Hill Cipher: As a polygraphic substitution cipher based on linear algebra, it hides single-letter frequencies better than Caesar, but it still exhibits strong linear patterns and poor diffusion. It fails to mimic a truly random sequence, thus failing most or all tests.
- DES & AES: Both are modern block ciphers designed with strict Confusion and Diffusion properties (the Avalanche effect). Even highly structured plaintext is mapped to ciphertext that is statistically indistinguishable from random noise. Therefore, both DES and AES are good, but only AES passed every single test, making it superior.

3.2

$\alpha = 0,05$	Frequency test	Serial test	Poker test	Runs test
LCG1	Fail	Fail	Pass	Pass
LCG2	Pass	Fail	Fail	Fail
LFSR1	Pass	Pass	Pass	Pass
LFSR2	Fail	Fail	Fail	Fail
BBS1	Pass	Pass	Fail	Fail
BBS2	Pass	Pass	Fail	Pass

Conclusions:

- Parameters dictate quality: The structure of a PRNG algorithm alone does not guarantee randomness. LCG and BBS can either perfectly pass or completely fail statistical testing strictly based on the parameters chosen.
- LCG constraints: LCG1 (Park-Miller) provides good statistical randomness, but LCG2 produces a trivially predictable alternating sequence because its modulus $m=2$ is far too small.
- LFSR polynomials: LFSR1 passes because a primitive polynomial ensures a maximal period. LFSR2 fails because non-primitive polynomials result in shortened periods and repeating bit patterns that are quickly caught by the Poker and Runs tests.
- BBS state space: The BBS algorithm is provably secure, but only if the modulus $M = p \times q$ is large enough to prevent short sequence loops. BBS1 fails due to a tiny modulus (77), while BBS2 behaves securely due to its larger prime parameters.